

What Is That Talk About? A Video-to-Text Summarization Dataset for Scientific Presentations

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Introduction

We propose **VISTA**, the first multimodal summarization dataset consisting of scientific presentation videos paired with paper abstracts.

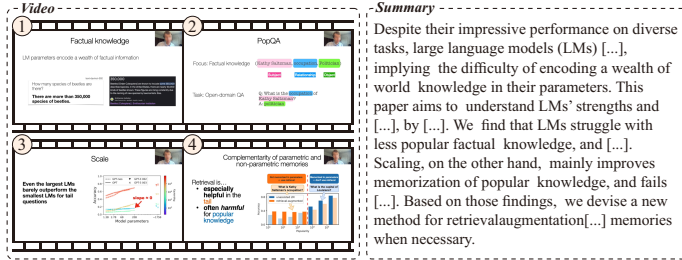


Figure 1. VISTA pairs presentation videos with paper abstracts

Plan-based Framework

- Problem:** SOTA LMMs show problems with structural grounding -> incoherence, hallucination
- Solution:** Introduce intermediate plan p as question sequence $\{q_1, q_2, \dots, q_m\}$
- Training:** Learn $P(s|v, p)$ (video v , summary s) instead of $P(s|v)$ mapping

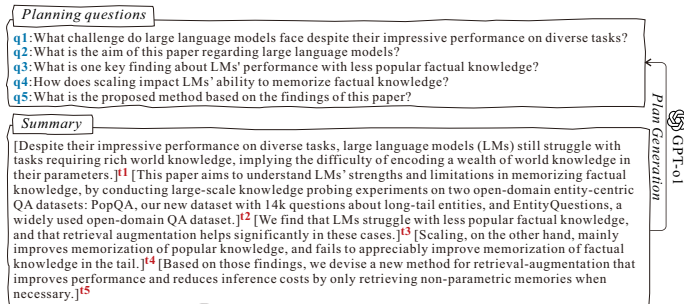


Figure 2. Plan extraction

The VISTA Dataset

- Scale:** 18,599 video-abstract pairs from leading AI conferences
- Sources:** ACL Anthology (ACL, EMNLP, NAACL, EACL), ICML, NeurIPS (2020-2024)
- Quality Control:** Manual validation (500 samples) + automated assessment (GPT-o1, All samples)
- Data Splits:** Train (80%), Validation (10%), Test (10%)

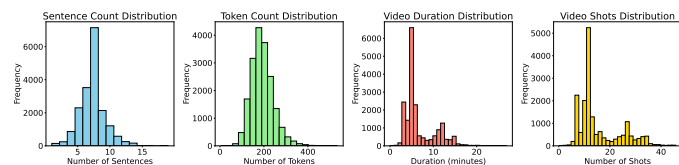


Figure 3. Dataset attribute distributions

- Videos:** Avg. 6.76 minutes, 16.36 shots per video
- Summaries:** Avg. 192.62 tokens, 7.19 sentences per summary
- Complexity:** Dependency tree depth 6.02, TTR 0.62

Main Results

- Plan-based superiority:** Planning model outperforms all baselines
- Modality ranking:** Video + Audio > Video > Audio > Transcript
- Modality interplay:** Video excels alone (rich cues), audio adds timing info, but transcripts are often noisy and hinder alignment
- Planning benefit:** Planning also boosts summarization for text- and audio-only models

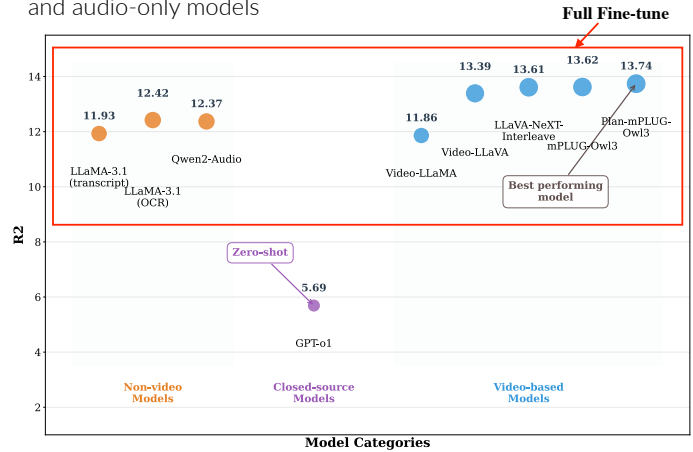


Figure 4. Model performance comparison

Human and GPT-o1 Evaluation

- Multi-aspect assessment:** Faithfulness, Relevance, Informativeness, Conciseness, Coherence
- Human superiority:** Humans consistently outperform all models across all evaluation criteria
- Plan-based advantage:** Plan-mPLUG-Owl3 achieves best performance among other models in both evaluations

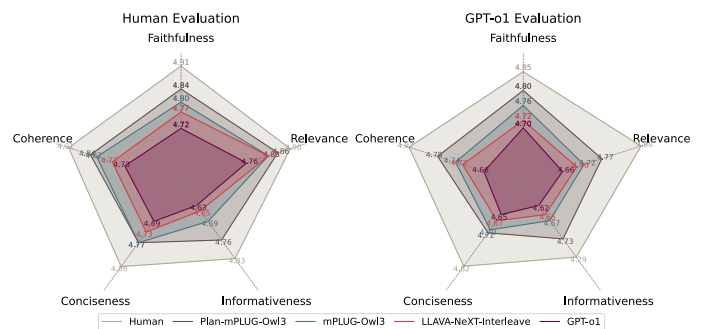


Figure 5. Human and GPT-o1 evaluation results

Conclusion

- Dataset:** VISTA provides 18,599 video-summary pairs, a novel large-scale dataset for scientific video-to-text summarization
- Benchmarking:** Comprehensive evaluation of 13+ SOTA LMMs across multiple settings (zero-shot, QLoRA, full fine-tuning)
- Method:** Plan-based summarization improves quality and factual accuracy over strong multimodal baselines

Project Info



dongqi.me/projects/VISTA